CHAPTER

3

DNA Quantitation



PURPOSE OF DNA QUANTITATION

When biological evidence from a crime scene is processed to isolate the DNA present, all sources of DNA are extracted and are present in the samples to be examined. Thus, non-human DNA, such as bacterial, fungal, plant, or animal material, may also be present in the total DNA recovered from the sample along with the human DNA of interest.

To ensure that DNA recovered from an extraction is human rather than from another source such as bacteria, the FBI's Quality Assurance Standards (Standard 9.4, QAS 2009) requires human-specific DNA quantitation. Only after DNA in a sample has been isolated can its quantity and quality be reliably assessed. The primary purpose of DNA quantitation in forensic casework is to determine the appropriate amount of DNA template to include in PCR amplification of short tandem repeat loci in order to avoid off-scale data and associated artifacts.

Determination of the amount of DNA in a sample is essential for most PCR-based assays because a narrow concentration range works best with multiplex short tandem repeat (STR) typing. Typically 0.5 ng to 2.0 ng of input human DNA is optimal with current commercial STR kits. With direct PCR (see Chapter 4), the number of PCR cycles can be adjusted so that a uniform spot of blood or buccal cells provides an equivalent amount of DNA and results in on-scale data.

PCR amplification of too much DNA results in overblown electropherograms that make interpretation of results more challenging and time-consuming to review. Too little DNA can result in loss of alleles due to stochastic amplification and failure to equally sample the STR alleles present in the sample. Figure 3.1 illustrates the "Goldilocks" principle of not wanting too much or too little DNA in a PCR reaction to produce an STR profile.

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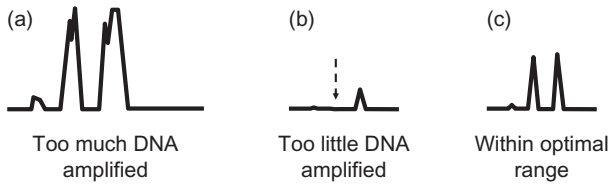


FIGURE 3.1 Illustration of STR typing results at a single heterozygous locus for a single source sample with (a) too much DNA template showing off-scale, split peaks, (b) too little DNA template where the arrow points to allele dropout due to stochastic effects, or (c) just the right amount so that two allele peaks are balanced and on-scale.

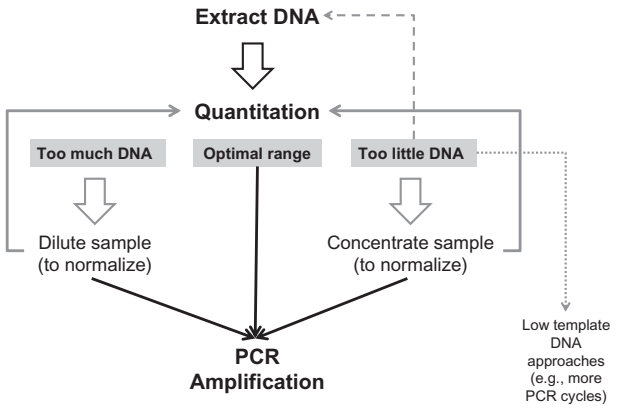


FIGURE 3.2 Flow chart illustrating role of DNA quantitation following DNA extraction. If the sample contains a DNA quantity within the optimal range, then an analyst would proceed with PCR amplification. With too much DNA, a sample could be diluted and either re-checked for DNA quantity or sent directly to the PCR amplification step. When DNA results below the optimal range are observed, the sample could be concentrated, re-extracted or treated with low template DNA approaches depending on laboratory policy and amount of available sample.

If the amount of DNA in a sample is outside of the target range for creating a "just right" DNA profile, then the DNA amount must be adjusted prior to putting it into the PCR reaction. The process of achieving a DNA concentration that fits the optimal window for analysis is called *normalization*. This involves diluting the sample down to the desired range or concentrating it by removing excess fluid. Figure 3.2 illustrates that if too little DNA is detected during the quantitation step then re-extraction or increased PCR cycles (provided the laboratory has a validated protocol, see Chapter 11) may be attempted.

Evaluation of human DNA quantity in a sample can be used to screen for samples that should be sent forward through the DNA testing process. When having to wade through a large number of samples, a sample screening process based on the amount of human DNA present can be very helpful and cost-effective. DNA quantitation that is performed well can save time during the data review process as data signal will be on-scale. By not having to repeat the testing process for an off-scale DNA result, the often-limited biological evidence will not have to be further consumed to try and obtain a better quality result.

TABLE 3.1 Quality Assessments Possible with Appropriate Quantitative PCR (qPCR) Assays.

Quality Assessment	How Assessed	Possible PCR Solutions
PCR inhibitor	IPC cycle threshold	PCR kit with improved buffer (Ch. 5)
Male-to-total DNA ratio is low	Dual Y and human qPCR	Y-STRs (Ch. 13)
Very low human genomic DNA	Multi-copy probe target can help; mtDNA can help	Increased cycles (Ch. 11); mtDNA (Ch. 14)
Degraded DNA	Different size qPCR targets	miniSTRs (Ch. 10); mtDNA (Ch. 14)
Non-human DNA	Species test (e.g., cyt b)	Use appropriate non-human DNA primers (Ch. 16)

In addition, DNA quantitation can serve as a gateway to potential DNA testing options. For example, if an assay can assess the relative levels of total genomic DNA compared to male DNA, then depending on the DNA quantitation results either autosomal STRs or Y-STRs may be attempted as a the first course of action with the evidentiary sample (Table 3.1).

DNA Quantities Used

PCR amplification is dependent on the quantity of template DNA molecules added to the reaction. Based on the amount of DNA determined to be in a sample with a quantitation method, the extracted DNA for each sample is adjusted to a level that will work optimally in the PCR amplification reaction. As mentioned above, commercial STR typing kits work best with an input DNA template of around 1 ng.

A quantity of 1 ng of human genomic DNA corresponds to approximately 303 copies of each locus that will be amplified (D.N.A. Box 3.1). There are approximately 6 pg (one millionth of one millionth of a gram or 10^{-12} grams) of genomic DNA in each cell containing a single diploid copy of the human genome. Thus, a range of typical DNA quantities from 0.1 ng to 25 ng would involve approximately 30 to 8330 copies of every nuclear DNA sequence to be examined.

Attempts to correlate the measured DNA quantity to PCR performance are complicated by the fact that target regions for the qPCR and STR assays are not the same. As more human genomes are being sequenced, we are learning that the differences between people can be greater than previously thought (e.g., copy number variation for large chromosomal regions).

DNA QUANTITATION METHODS

A number of DNA quantitation tests have been used over the years to estimate the amount of total DNA or human DNA present in a sample (Nicklas & Buel 2003, Barbisin & Shewale 2010). These DNA quantitation tests, which will be discussed briefly below, include UV absorbance, yield gels, slot blot, PicoGreen, end-point PCR, and real-time quantitative PCR. Early assays were "home-brew" (i.e., prepared by the laboratory performing the test)

D.N.A. BOX 3.1

CALCULATION OF DNA QUANTITIES IN GENOMIC DNA

1. Relative molecular mass of a DNA base pair = 618 g/mol
   A = 313 g/mol; T = 304 g/mol; A-T base pairs = 617 g/mol
   G = 329 g/mol; C = 289 g/mol; G-C base pairs = 618 g/mol
2. Relative molecular mass of DNA = 1.98×10^{12} g/mol
   There are 3.2 billion base pairs in a haploid cell ($\approx 3.2 \times 10^{9}$ bp).
   ($\approx 3.2 \times 10^{9}$ bp) × (618 g/mol/bp) = 1.98×10^{12} g/mol
3. Quantity of DNA in a haploid cell = 3 picograms
   1 mole = 6.02×10^{23} molecules
   (1.98×10^{12} g/mol) × (1 mole/6.02×10^{23} molecules)
   $= 3.3 \times 10^{-12}$ g = 3.3 picograms (pg)
   A diploid human cell contains ≈6.6 pg genomic DNA.
4. One ng of human DNA comes from ≈152 diploid cells.
   1 ng genomic DNA (1000 pg)/6.6 pg/cell = ≈303 copies of each locus (2 per 152 diploid genomes)

while most forensic DNA quantitation is now performed using commercial kits from suppliers like Applied Biosystems or Promega Corporation.

UV Absorbance and Yield Gels

Early methods for DNA quantitation typically involved either measurement of absorbance at a wavelength of 260 nm or fluorescence after staining a yield gel with ethidium bromide. Unfortunately, because these approaches are not very sensitive, they consume valuable and often irreplaceable forensic specimens. In addition, absorbance measurements are not specific for DNA, and contaminating proteins or phenol left over from the extraction procedure can give falsely high signals. To overcome these problems, several methods have been developed for DNA quantitation purposes. These include the slot blot procedure and fluorescence-based microtiter plate assays as well as so-called "real-time" or "quantitative PCR" approaches.

Slot Blot

The most commonly used method in forensic labs during the late 1990s and beginning years of the twenty-first century for genomic DNA quantitation was the so-called "slot blot" procedure. This test was specific for human and other primate DNA due to a 40 base pair probe that bound to a region on chromosome 17 called D17Z1. The slot blot assay was

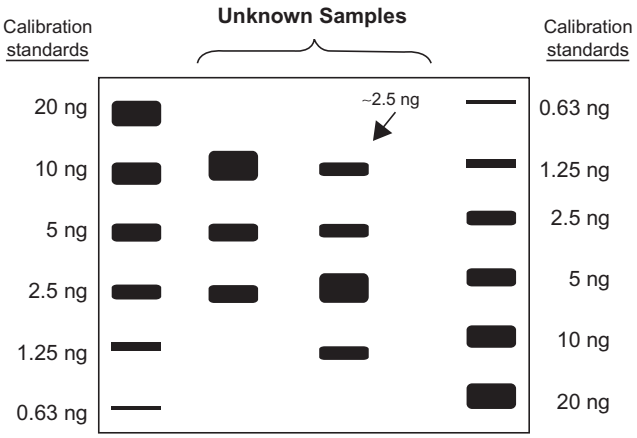


FIGURE 3.3 Illustration of a human DNA quantitation result with the slot blot procedure. A serial dilution of a human DNA standard is run on either side of the slot blot membrane for comparison purposes. The quantity of each of the unknown samples is estimated by visual comparison to the calibration standards. For example, the sample indicated by the arrow is closest in appearance to the 2.5 ng standard.

first described with radioactive probes (Waye et al. 1989), but was later modified and commercialized with chemiluminescent or colorimetric detection formats (Walsh et al. 1992, Budowle et al. 1995).

Slot blots involved the capture of genomic DNA on a nylon membrane followed by addition of a human-specific probe. Chemiluminescent or colorimetric signal intensities were then compared between a set of standards and the samples (Figure 3.3).

As with almost all DNA quantitation methods, the slot blot procedure involved a relative measurement that compared unknown samples to a set of standards. These standard samples were prepared via a serial dilution from a DNA sample of known concentration. The comparison between the standards and unknowns was usually performed visually—and therefore influenced by subjectivity of the analyst. However, digital capture and quantification of slot blot images was also an option using a charge-coupled device (CCD) camera imaging system (Budowle et al. 2001).

Up to 30 samples could be tested on a slot blot membrane with six to eight standard samples run on each side of the membrane for comparison purposes. For example, the standards might be a serial dilution of human DNA starting with 20 ng, 10 ng, 5 ng, 2.5 ng, etc. Typically about 5 μL of DNA extract from each sample was consumed in order to perform this DNA quantitation test.

The slot blot assay took several hours to perform and could detect both single-stranded and double-stranded DNA down to levels of approximately 150 pg (Walsh et al. 1992). Even when no results were seen with this hybridization assay, some forensic scientists still went forward with DNA testing and often obtained a successful STR profile. Thus, the slot blot assay was not as sensitive as desired. In addition, as with most "human-specific" tests, primate samples, such as chimpanzees and gorillas, also produced signal due to similarities in human and other primate DNA sequences.

In 2006, Applied Biosystems (Foster City, CA), the final commercial source for slot blot assay reagents, stopped selling the QuantiBlot Human DNA Quantitation Kit. Thus, this assay is now a thing of the past.

PicoGreen Microtiter Plate Assay

As higher throughput methods for DNA determination are being developed, more automated procedures are needed for rapid assessment of extracted DNA quantity prior to DNA amplification. To this end, in the mid-1990s, the Forensic Science Service (UK) developed a PicoGreen assay that is capable of detecting as little as 250 pg of double-stranded DNA in a 96-well microtiter plate format (Hopwood et al. 1997). PicoGreen is a fluorescent interchelating dye whose fluorescence is greatly enhanced when bound to double-stranded DNA (Ahn et al. 1996).

To perform this microtiter plate assay, 5 μL of sample are added to 195 μL of a solution containing the PicoGreen dye. Each sample is placed into an individual well on a 96-well plate and then examined with a fluorometer. A 96-well plate containing 80 individual samples and 16 calibration samples can be analyzed in less than 30 minutes (Hopwood et al. 1997). The DNA samples are quantified through comparison to a standard curve. This assay has been demonstrated to be useful for the adjustment of input DNA into the amplification reaction of STR multiplexes (Hopwood et al. 1997). It has been automated on a robotic workstation as well. Unfortunately, this assay quantifies total DNA in a sample and is not specific for human DNA.

AluQuant Human DNA Quantitation System

Around 2000, the Promega Corporation developed a human DNA quantitation system, known as AluQuant, that enabled fairly sensitive detection of DNA using *Alu* repeats that are in high abundance in the human genome (Mandrekar et al. 2001). Probe-target hybridization initiated a series of enzymatic reactions that ended in oxidation of luciferin with production of light. The light intensity was then read by a luminometer with the signal being proportional to the amount of DNA present in the sample. Sample quantities were determined by comparison to a standard curve. The AluQuant assay possesses a range of 0.1 ng to 50 ng for human DNA and can be automated on a robotic liquid-handling workstation (Hayn et al. 2004). While this assay was used for several years by laboratories such as the Virginia Department of Forensic Sciences (Greenspoon et al. 2006), it has been made obsolete with the introduction of real-time quantitative PCR (qPCR) assays.

End-Point PCR

A less elegant (and less expensive than qPCR, see below) approach for testing the "amplifiability" of a DNA sample is to perform an end-point PCR test. In this approach a single STR locus (Kihlgren et al. 1998, Fox et al. 2003) or other region of the human genome, such as an *Alu* repeat (Sifis et al. 2002, Nicklas & Buel 2003), is amplified along with DNA samples of known concentrations. A standard curve can be generated from the samples with known amounts to which samples of unknown concentration are compared.

A fluorescent intercalating dye such as SYBR Green (Zipper et al. 2004) can be used to detect the generated PCR products. Based on the signal intensities resulting from amplification of the single STR marker or *Alu* repeat region, the level of DNA can be adjusted prior to amplifying the multiplex set of DNA markers in order to obtain the optimal results. This method is a functional test because it also monitors the level of PCR inhibitors present in the sample. The amelogenin locus has also been used in conjunction with capillary electrophoresis separation and detection to provide gender and relative quantity assessment when compared to a standard curve of known DNA amounts (Allen & Fuller 2006).

In the end, each of the DNA quantitation methods described here has advantages and disadvantages and could be used depending on the equipment available and the needs of the laboratory.

REAL-TIME QUANTITATIVE PCR (QPCR)

The primary purpose in performing a DNA quantification test is to determine the amount of "amplifiable" DNA. A PCR amplification reaction may fail due to the presence of co-extracted inhibitors, highly degraded DNA, insufficient DNA quantity, or a combination of all of these factors. Thus, a test that can accurately reflect both the quality and the quantity of the DNA template present in an extracted sample is beneficial to making decisions about how to proceed. "Real-time" PCR assays provide such an assessment of both DNA yield and purity for amplification purposes.

Instruments and assays are now available that can monitor the PCR process as it is happening, enabling "real-time" data collection. Real-time quantitative PCR (qPCR), which was first described by Higuchi and co-workers at the Cetus Corporation in the early 1990s (Higuchi et al. 1992, 1993), is sometimes referred to as quantitative PCR or "kinetic analysis" because it analyzes the cycle-to-cycle change in fluorescence signal resulting from amplification of a target sequence during PCR. This analysis is performed without opening the PCR tube and therefore can be referred to as a closed-tube or homogeneous detection assay.

Several approaches to performing qPCR homogeneous detection assays have been published. The most common approaches utilize either the fluorogenic 5′ nuclease assay—better known as Taqman—or use of an intercalating dye, such as SYBR Green, that is highly specific for double-stranded DNA molecules. The TaqMan approach monitors change in fluorescence due to displacement of a dual dye-labeled probe from a specific sequence within the target region while the SYBR Green assay detects formation of any PCR product.

Instruments Used For qPCR

A number of different instruments have been used for qPCR assays and commercial kits include the ABI 7000 (Applied Biosystems, Foster City, CA), ABI 7500, ABI 7700, ABI 7900, the Corbett Rotogene 3000 (now sold as the QIAGEN Roto-Gene Q by QIAGEN, Valencia, CA), the iCycler (Bio-Rad, Hercules, CA), and the Roche LightCycler (Roche Applied Science, Indianapolis, IN). Currently, the most widely used qPCR instrument in forensic DNA laboratories is probably the ABI 7500.

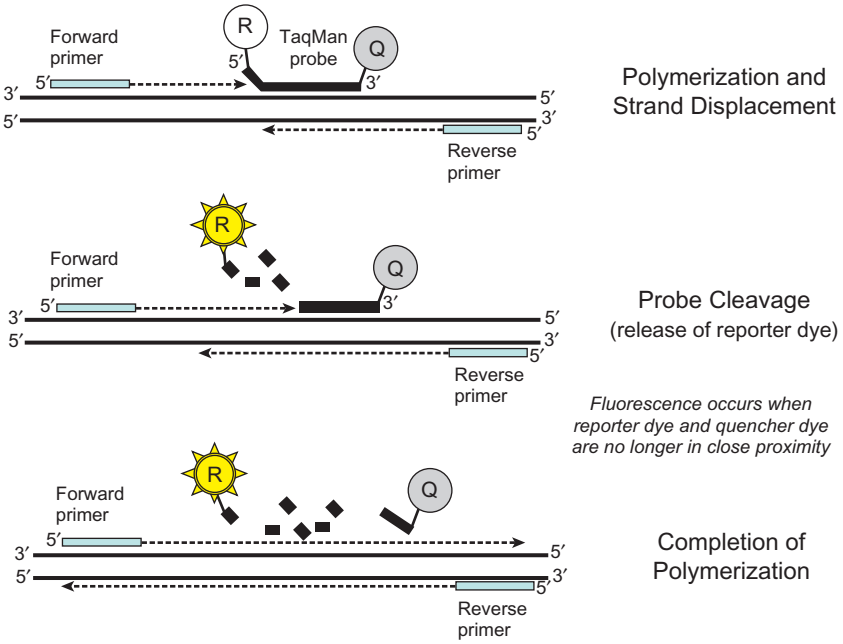


FIGURE 3.4 Schematic of TaqMan (5′ nuclease) assay.

The 5′ Nuclease Assay (TaqMan)

TaqMan probes are labeled with two fluorescent dyes that emit at different wavelengths (Figure 3.4). The probe sequence is intended to hybridize specifically in the DNA target region of interest between the two PCR primers. Typically the probe is designed to have a slightly higher annealing temperature compared to the PCR primers so that the probe will be hybridized when extension (polymerization) of the primers begins. A minor groove binder is sometimes used near the 3′-end of TaqMan probes to enable the use of shorter sequences that have higher annealing temperatures than would be expected for sequences of equivalent length.

The "reporter" (R) dye is attached at the 5′-end of the probe sequence while the "quencher" (Q) dye is synthesized on the 3′-end. A popular combination of dyes is FAM or VIC for the reporter dye and TAMRA for the quencher dye. When the probe is intact and the reporter dye is in close proximity to the quencher dye, little to no fluorescence will result because of suppression of the reporter fluorescence due to an energy transfer between the two dyes.

During polymerization, strand synthesis will begin to displace any TaqMan probes that have hybridized to the target sequence. The Taq DNA polymerase used has a 5′-exonuclease activity and therefore will begin to chew away at any sequences in its path (i.e., those probes that have annealed to the target sequence). When the reporter dye molecule is released from the probe and is no longer in close proximity to the quencher dye, it can begin to fluoresce (Figure 3.4). Increase in the fluorescent signal results if the target sequence is

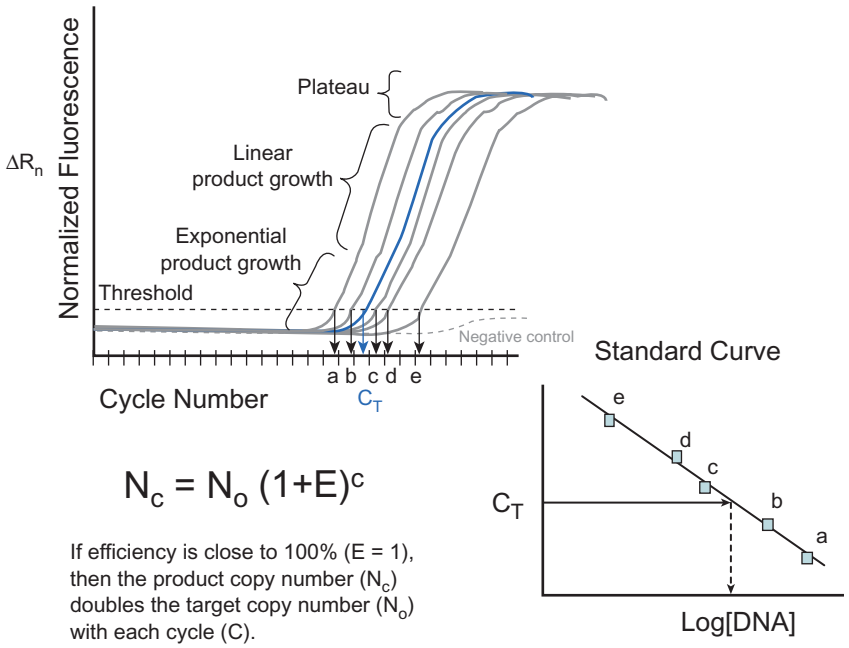


FIGURE 3.5 Real-time PCR output and example standard curve used to determine quantity of input DNA.

complementary to the TaqMan probe. It is important to note that mismatches between the DNA template sequence and the TaqMan probe can cause failure to detect the DNA template appropriately (Smith et al. 2002).

Some assays, such as the Quantifiler kit, include an internal PCR control (IPC) that enables verification that the polymerase, the assay, and the detection instrumentation are working correctly. In this case, the IPC is labeled with a VIC (green) reporter dye and hybridizes to a synthetic template added to each reaction. The TaqMan probe for detecting the target region of interest is labeled with a FAM (blue) reporter dye and is therefore spectrally resolvable from the green VIC dye. Instruments such as the ABI Prism 7000 or 7500 Real Time PCR System enable another dye like ROX (red dye) to be placed in each well to adjust for well-to-well differences across a plate through background subtraction.

Real-Time PCR Analysis

There are three distinct phases that define the PCR process: geometric or exponential amplification, linear amplification, and the plateau region. These regions can be seen in a plot of fluorescence versus PCR cycle number (Figure 3.5). During exponential amplification, there is a high degree of precision surrounding the production of new PCR products. When the reaction is performing at close to 100% efficiency, then a doubling of amplicons occurs with each cycle. A plot of cycle number versus a log scale of the DNA concentration should result in a linear relationship during the exponential phase of PCR amplification.

A linear phase of amplification follows the exponential phase as one or more components fall below a critical concentration and amplification efficiency slows down to an arithmetic increase rather than the geometric one in the exponential phase. Since components such as deoxynucleotide triphosphates (dNTPs) or primers may be used up at slightly different rates between reactions, the linear phase is not as consistent from sample-to-sample and therefore is not as useful for comparison purposes.

The final phase of PCR is the plateau region where accumulation of PCR product slows to a halt as multiple components have reached the end of their effectiveness in the assay. The fluorescent signal observed in the plateau phase levels out. The accumulation of PCR product generally ceases when its concentration reaches approximately 10^{-7} mol/L.

The optimal place to measure fluorescence versus cycle number is in the exponential phase of PCR where the relationship between the amount of product and input DNA is more likely to be consistent. Real-time PCR instruments use what is termed the cycle threshold (C_T) for calculations. The C_T value is the point in terms of PCR amplification cycles when the level of fluorescence exceeds some arbitrary threshold, such as 0.2, that is set by the real-time PCR software to be above the baseline noise observed in the early stages of PCR. The fewer cycles it takes to get to a detectable level of fluorescence (i.e., to cross the threshold set by the software), the greater the initial number of DNA molecules put into the PCR reaction. Thus a plot of the log of DNA concentrations versus the C_T value for each sample results in a linear relationship with a negative slope (Figure 3.5).

The cleavage of TaqMan probes or binding of SYBR Green intercalating dye to double-stranded DNA molecules results in an increase in fluorescence signal. This rise in fluorescence can be correlated to the initial DNA template amounts when compared with samples of known DNA concentration. For example in Figure 3.5, five samples (a,b,c,d,e) are used to generate a standard curve based on their measured C_T values. Provided that the PCR amplification conditions are consistent from sample-to-sample, a sample with an unknown DNA quantity can be compared to this standard curve to calculate its initial DNA template concentration.

In October 2007, the U.S. National Institute of Standards and Technology (NIST) released a reference material for human DNA quantitation. Standard Reference Material (SRM) 2372 contains three DNA samples that can be used to calibrate human DNA quantity measurements (D.N.A. Box 3.2).

Several real-time PCR assays have been developed with the human identity testing market in mind. Commercial kits for detecting human DNA as well as a real-time PCR assay for determining the amount of human Y-chromosome DNA present in a sample are now available. These kits include Quantifiler, Quantifiler Y, and Quantifiler Duo from Applied Biosystems and Plexor HY from Promega Corporation (Table 3.2).

Interlaboratory Studies

Several interlaboratory tests to evaluate DNA quantification methods have been conducted by NIST to better understand the measurement variability seen with various techniques (Duewer et al. 2001, Kline et al. 2003). A ten-fold range of reported concentrations was observed in one study (Figure 3.6). Most DNA quantitation measurements are precise to within a factor of two if performed properly (Kline et al. 2003, Kline et al. 2005). While

D.N.A. BOX 3.2

NIST SRM 2372: HUMAN DNA QUANTITATION STANDARD

In October 2007, after many years of research, the U.S. National Institute of Standards and Technology (NIST) released Standard Reference Material (SRM) 2372 for quantitation of human genomic DNA (Kline et al. 2009, Vallone et al. 2008). SRM 2372 contains three components labeled A, B, and C. All components are human genomic DNA. Component A was prepared at NIST from Buffy coat white blood cells from a single source anonymous male. Component B was prepared at NIST from Buffy coat cells from multiple anonymous female donors. Component C was obtained as a commercial lyophilized human genomic extract and has both male and female donors.

The DNA materials were characterized based on their absorbance properties at 260nm. Under the assumption that an absorbance of 1.0 at 260nm equals 50ng/μL of double-stranded DNA, nominal DNA concentrations of 52.5, 53.6, and 54.3ng/μL are present in the SRM 2372 components. These materials were initially characterized with Quantifiler and Quantifiler Y kits (Green et al. 2005), the "Vermont's *Alu*" assay (Nicklas & Buel 2003), and the CFS-HumRT assay (Richard et al. 2003). Subsequent studies of the three SRM 2372 components (LaSalle et al. 2011) with Quantifiler Duo (Barbisin et al. 2009) and Plexor HY (Krenke et al. 2008), which became available after the release of SRM 2372, have shown that there are some apparent copy number effects with these kits. Thus, it is important to remember that reference materials can only be held to their certified values and may not be fully computable to future assays or commercial kits.

SRM 2372 is not intended for use as a daily calibrant with qPCR assays. Rather,

DNA standard calibration using SRM 2372

	Commercial standard							
	1		2		3		4	
	[DNA]	S.D.	[DNA]	S.D.	[DNA]	S.D.	[DNA]	S.D.
Dilution								
10×	105	3.2	122	1.0	126	5.8	256	10.1
50×	105	3.3	122	7.3	145	0.8	272	7.8
100×	99	6.2	113	11.6	138	0.5	270	10.5
200×	100	1.7	137	18.5	137	3.9	311	3.7
Assigned	102		123		136		277	
Stated	200		200		200		260	
Deviation (%)	−49		−38		−32		6	

D.N.A. BOX 3.2 (*cont'd*)

laboratories can value assign their individual kit/assay calibrant materials based on comparison to SRM 2372. This may be done by diluting a kit calibrant or commercial DNA standard multiple times and testing it using dilutions of an SRM 2372 component for the standard curve. As noted in Vallone et al. (2008) Table 2 (which is reproduced at the bottom of the previous page), different lots of a commercial calibrant (samples 1, 2, and 3 in the Vallone et al. 2008 Table 2 example) may differ by more than 30% from their stated value of 200 ng/μL (e.g., 102, 123, and 136 ng/μL). Assigning these samples their appropriate values (e.g., 102 ng/μL instead of 200 ng/μL) will enable consistency over time and better correlation of DNA quantitation values with STR typing results.

TABLE 3.2 Real-Time Quantitative PCR Assays for DNA Quantitation and Assessment. For a More Complete List, See Barbisin & Shewale (2010).

Assay/Kit	Chemistry	Application	Target Gene	Amplicon Size	Standard Curve Range
Quantifiler	TaqMan	Total human DNA	Human: telomerase reverse transcriptase (hTERT) IPC: synthetic oligo	62 bp 79 bp	46 pg to 100 ng
Quantifiler Y	TaqMan	Human male DNA	Male: sex-determining region Y chromosome (SRY) IPC: synthetic oligo	64 bp 79 bp	46 pg to 100 ng
Quantifiler Duo	TaqMan	Total human & male DNA	Human: ribonuclease P RNA component 1 (RPPH1) Male: SRY	140 bp 130 bp	46 pg to 100 ng
Plexor HY	Plexor	Total human & male DNA	Human: RNU2 (multi-copy) Male: TSPY (multi-copy)	99 bp 133 bp	6.4 pg to 100 ng
"Vermont's" *Alu* (Nicklas & Buel 2003)	SYBR	Total human DNA	Human: Ya5 subfamily *Alu* repeat (multi-copy)	124 bp	1 pg to 16 ng
"CFS" HumRT (Richard et al. 2003)	TaqMan	Total human DNA	Human: flanking region of TH01 STR locus	62 bp	25 pg to 25 ng
CA DOJ quadruplex (Hudlow et al. 2008)	TaqMan	Total human, degradation, & male DNA	Human: TH01 STR allele Male: SRY Degradation: CSF1PO flanking IPC: synthetic oligo	≈170–190 bp 137 bp 67 bp 77 bp	7.2 pg to 100 ng

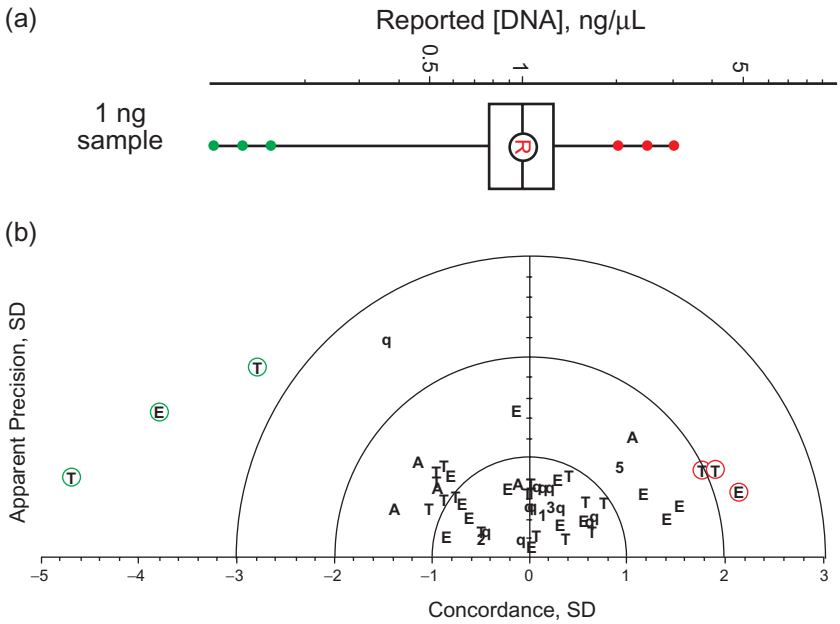


FIGURE 3.6 (a) Range of DNA concentrations reported for a 1 ng DNA sample supplied to 74 laboratories in an interlaboratory study (Kline et al. 2003). Overall the median value was very close to the expected 1 ng level with 50% falling in the boxed region. However, laboratories returned values ranging from 0.1 ng to 3 ng. (b) A target plot examining concordance and apparent precision for the various laboratory methods used. Legend: A = ACES kit; q = Quantiblot with unreported visualization method; E = Quantiblot with chemiluminescent detection; T = Quantiblot with colorimetric detection; 1, 2, 3, 4, and 5 represent methods used by only one lab.

this degree of imprecision may seem large, recall that a factor of two corresponds to one exponential-phase PCR amplification cycle; quantitation results are usually sufficiently valid to estimate DNA template amounts that will enable optimal PCR amplification.

In the NIST Quantitation Study 2004 (QS04), a total of 60 data sets from 287 submitted involved qPCR (Kline et al. 2005). Of the 60 qPCR data sets, 37 came from Quantifiler (Figure 3.7). Overall the Quantifiler assay performed well with the median value from participants coming close to the expected value (center of the target plot in Figure 3.7). However, outliers did exist emphasizing the need for care in pipetting and conducting qPCR assays.

Correlation of DNA Quantity and STR Amplification

In spite of the sensitivity of qPCR, some studies have shown that STR typing results can be obtained even when a "zero" quantitative value is observed (Cupples et al. 2009). Stochastic variation with low amounts of DNA is the reason for such observations (see Chapter 4). While there is DNA present in such samples, the qPCR result is very low or zero due to the PCR primers failing to find sufficient target to amplify. How then can qPCR

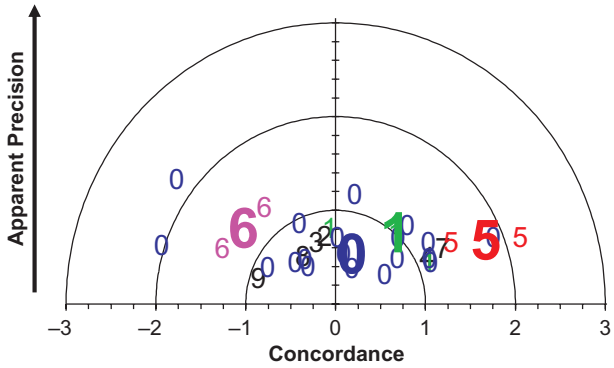


FIGURE 3.7 Target plot comparing interlaboratory results from 60 data sets involving 8 different samples using 10 different qPCR methods (Kline et al. 2005). Larger characters in bold font represent the median performance for all results submitted for a particular method. Note that specific methods may exhibit a bias relative to other assays. 0 = Quantifiler, which represented 37 of the 60 data sets (see Table 1 in Kline et al. 2005 for full description of qPCR assay codes).

results be an appropriate gatekeeper about whether or not a DNA sample should be processed further?

It is important that several things be kept in mind. First, just as with STR typing results from low amounts of DNA, stochastic variation can limit the reliability of qPCR results due to allele dropout. And, as with low template DNA testing (see Chapter 11), replicate qPCR testing is a possible solution to strengthen confidence in the result (D.N.A. Box 3.3).

Second, remember that there are different volumes of input DNA being used. Many qPCR assays require 2 µL of input DNA while STR typing PCR reactions can take 10 µL or more of input DNA. Thus, because five times as much DNA extraction volume can be included in the STR amplification reaction, more input DNA can be included giving rise to a result when the qPCR value was "zero."

Third, the PCR buffers between the qPCR and STR reactions may be different. If the STR amplification buffer contains a different polymerase or materials to enable overcoming PCR inhibition, then results may not be equivalent and the qPCR assay may not provide a true measure of STR typing performance. Furthermore, the different input volumes going into the qPCR versus STR amplification reactions could lead to different concentrations for PCR inhibitors coming from casework samples so that the qPCR or STR amplifications fail at a different rate.

Fourth, pipetting accuracy may be a factor. Pipetting 2 µL is generally less accurate than is pipetting 10 µL. A mis-pipetting of the DNA sample going into the qPCR assay could make a result appear lower than it really is. Reduction of volume to save money with qPCR assays (Westring et al. 2007) could exacerbate pipetting accuracy issues as well as effectively concentrate PCR inhibitors.

Finally, it is important that the qPCR result is appropriately correlated with STR typing performance. Internal validation (see Chapter 7) is crucial in developing appropriate interpretation of results. Variation in different lots of qPCR kit calibrants have led to problems in

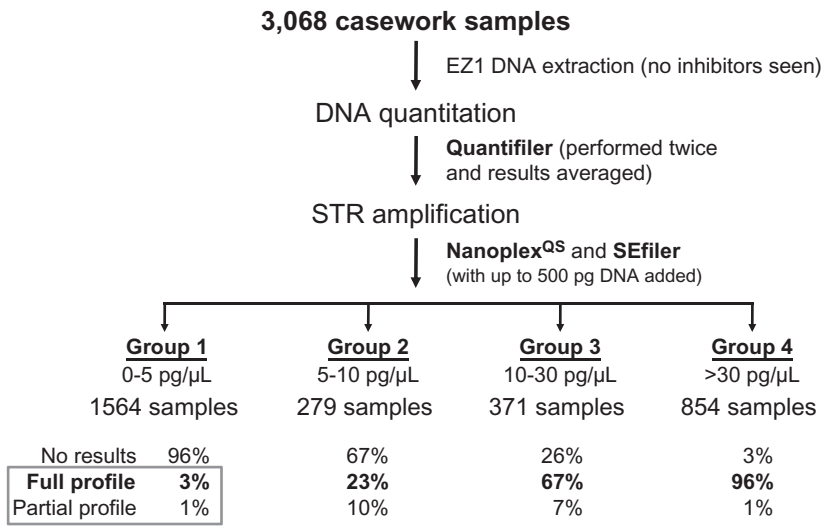
D.N.A. BOX 3.3

REPLICATE TESTING AIDS CONFIDENCE IN DNA QUANTITATION VALUES OF "ZERO"

A study by the Institute of Legal Medicine in Munich, Germany attempted to correlate the DNA quantitation values with STR performance (Kremser et al. 2009). A set of 3,068 casework samples that had been extracted using the QIAGEN EZ1 robot was tested twice with the Quantifiler qPCR kit and the results averaged. Based on the internal positive control (IPC) during qPCR showing no increase in cycle threshold values, it was assumed that potential PCR inhibitors had been removed during DNA extraction. STR amplification was then performed with the Nanoplex^{QS} kit from Biotype (Dresden, Germany) followed by the SEfiler kit (Applied Biosystems) to confirm allele calls on all positive results. Based on the average of the two Quantifiler results, samples were divided into four groups: Group 1 (0 pg/μL to 5 pg/μL), Group 2 (5 pg/μL to 10 pg/μL), Group 3 (10 pg/μL to 30 pg/μL), and Group 4 (>30 pg/μL).

Generally, STR typing results correlated with the amount of DNA. Full profiles were observed 96% of the time when >30 pg/μL were reported. Likewise, no STR results were obtained 96% of the time when DNA quantities in the range of 0 pg/μL to 5 pg/uL were reported. However, full or partial DNA profiles were observed 4% of the time when essentially the DNA quantity was zero (Group 1, 0 pg/μL to 5 pg/uL obtained). The 1564 samples in Group 1 were explored further by examining the individual Quantifiler results.

When both Quantifiler replicate results were zero, then 100% of the time subsequent STR typing failed to obtain results (from 750 tested samples). Thus, when using DNA quantitation in a gatekeeper function for whether or not to proceed with further



D.N.A. BOX 3.3 (*cont'd*)

1564 Samples with "Zero" Quantifiler Results (pg/μL)

	0,0	0,>0	>0,>0
Number of Samples	750	478	336
Positive results	0%	7%	27%
Negative results	100%	93%	63%

testing, replicate zero values were important to guarantee no DNA was present. This same concept of replicate testing aids reliability with low template DNA (see Chapter 11).

***Source*:**

Kremser, A., et al. (2009). Quantifiler Human DNA Quantification Kit (Applied Biosystems) as a screening kit for DNA profiling. Forensic Science International: Genetics Supplement Series, *2, 106–107*.

correlating DNA quantitation values with expected STR typing performance (Nielsen et al. 2008). Thus, calibration to a certified reference material such as NIST SRM 2372 (D.N.A. Box 3.2) is valuable in adjusting qPCR kit calibrant material amounts in order to maintain consistency in DNA quantitation results over time.

DNA QUANTITATION AS A GATEKEEPER

While DNA quantitation is not mandatory (except for casework samples as described under FBI Quality Assurance Standard 9.4), it is a good idea and aids adjustment of DNA input amounts into PCR to enable on-scale STR allele peaks for improved data interpretation. However, DNA quantity measurements are usually not perfect and represent a "ballpark figure." Internal validation to correlate qPCR values to expected STR results is crucial.

Moreover, if new STR kits with improved buffer formulations to overcome PCR inhibitors are used (e.g., PP16 HS and Identifiler Plus, see Chapter 5), then many qPCR assays (e.g., Quantifiler) may no longer be an appropriate gatekeeper. Thus, qPCR DNA quantitation systems must be compatible with new and improved STR systems.

A properly tuned and applied DNA quantity/quality assay can save time and expense in terms of halting downstream sample processing when little-to-no DNA is present. Too avoid problems associated with low-level DNA testing (see Chapter 11), a laboratory may opt to halt testing for a specific sample when quantitation results below a specified validated threshold are observed for that sample. However, as noted in D.N.A. Box 3.3, replicate qPCR testing is advisable to strengthen reliability in correlating DNA quantitation results with expected STR typing performance.

READING LIST AND INTERNET RESOURCES

General Information

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